

Temporal Directedness Without an Open Future

Tenseless modal structure and the orientation of a block universe

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Abstract: How can time have a physical direction if the whole history of the universe already exists? This paper gives a conditional account of temporal orientation through a rooted modal structure and its physical projection. Two complementary results establish the argument. Proposition 1 shows that a rule respecting exactly reversal-symmetric explanatory data cannot distinguish one time orientation from its reverse. Proposition 2 gives sufficient conditions for a rooted order of admissible event extensions to provide the required asymmetry. Its physical projection must yield at least one non-zero outgoing history-direction; every such direction must be timelike, and all must agree with one global orientation. That orientation is then uniquely compatible with the extension order on a connected, time-orientable spacetime. A Janus-style example shows how one global orientation can coexist with opposing local entropy arrows, and why the bridge fails when projected extensions demand opposite orientations. The contribution is an explicit connection between modal extension and physical orientation in a complete tenseless block. The extension order and projection are independently specified, and the result is conditional on their physical compatibility. Explaining thermodynamic irreversibility requires a further account of preparation and dynamics.

Keywords: eternalism; branching space-times; temporal orientation; modality; block universe; time reversal

1 Introduction: what needs explaining?

There are several questions hidden under the expression 'the direction of time'. One can ask whether spacetime is time-orientable, what physically distinguishes one of its two possible orientations, and why irreversible processes such as entropy increase align with one orientation. Earman (1974) already insisted that these questions should be disentangled. Farr's discussion of C-theories makes the same lesson vivid from another direction: whether time itself has a direction is a substantive question, not a verbal consequence of saying that the universe expands or that entropy increases (Farr 2020).

This paper asks what can physically distinguish one of the two time orientations of a tenseless block without defining the future by entropy increase. The difficulty is specific. A time-orientable spacetime always admits the reversed orientation as well. If the data admitted into an explanation are unchanged when the two orientations are exchanged, nothing in those data can say which orientation is the future (Section 2). An explanation must therefore add an asymmetric structure and show how it fixes one orientation. That structure can be a primitive direction, an asymmetric boundary condition, a global geometric asymmetry or, as proposed here, an asymmetric modal structure.

Two familiar ideas pull in different directions. Eternalism treats events at different times as equally real and is naturally represented by a complete tenseless block. Branching approaches make objective modality central. In Branching Space-Times, histories represent alternative possible courses of events and can overlap before diverging (Belnap 1992; Belnap,

Müller, and Placek 2021). Branching therefore seems to offer resources for directedness that a bare symmetric block lacks.

The price is often taken to be an ontologically open future, which would conflict with eternalism. McCall's branching universe ties temporal direction to a tree of possibilities and branch attrition (McCall 1994). Standard branching semantics implements a fixed-past condition, and Placek develops a modal introduction of past, present and future (Placek 2021). Farr, however, argues that McCall's attrition and the no-backward-branching feature of branching spacetimes import substantive A- and B-theoretic commitments (Farr 2012).

Explicit eternalist readings of branching time already exist (Spolaore and Zanardo 2025), and branching semantics can be formulated tenselessly. In standard branching models, a fixed-past or no-backward-branching condition gives extension its temporal reading (Farr 2012). The question addressed here is how an extension order specified independently of a time orientation can acquire physical significance. Proposition 1 identifies the obstruction posed by reversal-symmetric explanatory data. Proposition 2 supplies a conditional bridge from an asymmetric modal order to a unique spacetime orientation. The Janus example tests this bridge and separates its result from the thermodynamic arrow (Section 7).

The construction begins with complete histories and their finite configurations. Starting from a minimal root, configurations are ordered by admissible extension: an extension retains its antecedent events and adds compatible events. This order is defined independently of a time orientation of spacetime. A separate projection Π maps its extension directions into spacetime. Timelike directions are tangent directions available to a massive particle's trajectory. If at least one projected history-direction is non-zero, and every such direction is timelike and agrees with one global time orientation, that orientation is uniquely compatible with the extension order. Opposing projected directions prevent a global choice. Orientation selection here concerns which temporal direction agrees with the modal structure; actuality concerns which complete history is realised.

The proposal competes with other strategies. Maudlin is prepared to take temporal direction or passage as fundamental (Maudlin 2007). Castagnino and Lombardi derive a non-entropic arrow from a global geometrical asymmetry of spacetime and transfer it to local energy flow (Castagnino and Lombardi 2009). I do not claim that modal structure is the only possible source. The narrower claim is that the explanatory resource associated with branching does not require an ontologically open future, and that a single modal primitive can do work beyond merely replacing a primitive temporal arrow.

2 A restricted symmetry no-go

Proposition 1 identifies the obstruction: explanatory data unchanged by reversal leave the two orientations indistinguishable to a rule respecting that symmetry. An explanation must therefore supply an asymmetric resource. In the present construction, that resource is the rooted order from antecedent configurations to their admissible extensions. Proposition 2 then states

when its physical projection distinguishes a unique spacetime orientation. The two propositions establish successive steps of the argument.

2.1 *Time-orientability is not yet a physical choice*

Let M be a connected time-orientable Lorentzian spacetime. Time-orientability means that the timelike cones admit a continuous division into two global components. It does not by itself tell us which component is physically future-directed. Once one orientation is available, its reverse is available as well.

A realistic cosmological block may already contain asymmetric physical features: an entropy gradient, expansion, radiation conditions, asymmetric matter distributions, or a special boundary. If those are admitted as inputs, the relevant data need not be reversal-invariant. The following proposition therefore has a deliberately restricted target: can a preferred orientation be extracted from data that themselves remain invariant under the reversal exchanging the two candidate orientations?

Let D be the admissible data from which an orientation rule is allowed to infer an answer. Let $O(M) = \{o+, o-\}$. An orientation rule is intrinsic relative to D when its output is a function of D alone, and equivariant when symmetries of D act correspondingly on its output.

2.2 *Proposition 1: reversal-symmetry no-go*

Proposition 1. Suppose there is an automorphism Θ of D such that $\Theta D = D$ and Θ exchanges $o+$ and $o-$. Let F be an intrinsic equivariant rule with values in $\{o+, o-, \text{undetermined}\}$. Then $F(D)$ cannot be exactly one of $o+$ or $o-$.

In plain language, identical reversal-symmetric input cannot tell an intrinsic symmetry-respecting rule which end is future.

Proof. Since $\Theta D = D$, $F(\Theta D) = F(D)$. Equivariance gives $F(\Theta D) = \Theta F(D)$. Hence $F(D) = \Theta F(D)$. Neither $o+$ nor $o-$ is fixed by Θ . Therefore a unique orientation cannot be selected from D alone. $\square$

What the proof establishes: This is a restricted impossibility result. A genuinely asymmetric boundary or other admitted data can break the symmetry.

This is a Curie-style symmetry obstruction rather than a deep new theorem. Curie's principle tells us to look for an asymmetry in the explanatory basis of an asymmetric effect; Earman (2004) shows both the usefulness and limitations of that principle. Here its role is diagnostic. If a block is reversal-symmetric relative to the admitted data, then an explanation of a preferred orientation must either enlarge those data, take the orientation as primitive, or invoke symmetry breaking.

The restriction is essential. A block containing a low-entropy boundary at one end is not invariant under reversal when that boundary is included in D . An entropy-based orientation therefore escapes Proposition 1. The present question is different: can one orient a block without using the contingent thermodynamic asymmetry as the orienting primitive? This

matters if one later wants to explain why entropy increase aligns with an independently fixed temporal orientation rather than define the orientation by entropy increase.

3 What branching contributes, and what it costs

The no-go result creates a search problem: what independently motivated structure could supply the required asymmetry? Branching theories offer a candidate because their possibilities are ordered, but standard interpretations often tie that order to a settled past and an open future. This section keeps the explanatory resource while identifying the temporal assumptions that must be removed.

Branching theories provide an asymmetric modal structure. Branching Space-Times represents objective indeterminism by a structured set of histories rather than by a mere ensemble of unrelated models (Belnap 1992; Placek and Belnap 2012). Histories can share events and diverge, and the overlap structure carries modal information.

The standard temporal reading exploits a fixed-past or no-backward-branching condition. Histories that agree at a later point agree at earlier points; alternatives open in one direction. Placek (2021) uses modal differences of this kind to introduce temporal notions. McCall (1994) makes the relation between branching and temporal direction especially explicit through branch attrition.

Farr (2012) is a useful warning against treating this asymmetry as innocent. He argues that McCall's attrition requires extra temporal structure and that the no-backward-branching condition in Belnap-style models can impose a time-asymmetric modal structure without sufficient physical justification. Branching can explain directedness only if its own directedness is understood.

This is where the present proposal departs from the open-future strategy. I do not interpret extension as 'what is not yet settled'. The histories can all be complete. The modal asymmetry is an order among complete possibilities, not an ontological difference between an existing past and a not-yet-existing future.

Nor is the mere existence of an eternalist branching semantics enough. Eternalist readings of branching time can formulate the semantics tenselessly (Spolaore and Zanardo 2025) while retaining a fixed-past condition. Such models establish compatibility between eternalism and branching, but they do not by themselves explain how a non-temporal modal order selects a physical time orientation. That is the bridge problem addressed below. The next section therefore reconstructs the branching asymmetry without an open future before Section 5 asks whether it can orient the physical block.

4 Tenseless extension among complete histories

The construction separates modal extension from temporal becoming. A complete history contains a coherent set of events, and smaller configurations describe compatible

portions of it. The order from antecedent configurations to their extensions can be asymmetric even though no history grows in an external time. The question is whether this structural asymmetry can orient spacetime through a separately specified physical projection.

Let E be a structured domain of admissible histories. A history H is complete: it is a maximal coherent configuration, not an object that literally grows in an external meta-time. This is the eternalist component of the proposal.

Finite configurations in E can nevertheless stand in an extension relation. Write $C < D$ when D is an admissible extension of C that preserves the relevant antecedent information. Assume a distinguished minimal configuration r . A chain has the schematic form $r < C_1 < C_2 < \dots$.

The asymmetry lies in the minimal root and the distinction between antecedents and extensions. Together they give the modal structure a primitive direction. Complete histories retain their tenseless status: the extension relation orders their configurations. Its connection to physical time is determined by the projection conditions introduced in Section 5.

Branching occurs when a configuration has incompatible extensions. Independent events can also produce diamonds: extending by a and then b can reach the same configuration as extending by b and then a . The relevant object is therefore not necessarily a strict tree. What matters is a rooted, globally coherent extension order that preserves provenance when histories are genuinely incompatible.

This separation is conceptually important. In standard temporal branching it is natural to read the direction of extension as future-directed from the start. Here that identification is withheld. The order in E is defined without using a time orientation of M . Whether it aligns with one is a further physical question. Section 5 turns that question into a compatibility test between E and M .

5 The orientation bridge

The projection Π connects modal extension to physical directions in spacetime. Proposition 2 asks whether all non-zero projected history-directions agree with one of the two available time orientations. It supplies the constructive step after Proposition 1: the enriched structure can distinguish an orientation precisely when the stated compatibility conditions hold.

Let Π map the regular event structure of E into a connected time-orientable Lorentzian block M . Some directions in E may be purely modal: they distinguish alternatives while projecting to the same event. Other directions are history-directions: their projections move non-trivially in M . Only the latter can carry temporal information.

Assume that relevant paths in E admit differentiable representatives and that Π is differentiable along them. For an outgoing history-direction v at e , $d\Pi_e(v)$ is a tangent vector at $x=\Pi(e)$. Let S be the set of points x for which at least one non-zero projected outgoing history-direction is available.

M is assumed time-orientable independently. It therefore admits two opposite global time orientations, τ and $-\tau$. We ask whether the projected directions are compatible with one of these orientations but not the other.

5.1 *Proposition 2: orientation selection by projected extension*

Proposition 2. Let M be connected and time-orientable, and let S be non-empty. Suppose every non-zero projected outgoing history-direction at every $x \in S$ is timelike. If there exists a global time orientation τ of M with respect to which all these projected directions are future-directed, then τ is the unique time orientation of M compatible with the projected extension order.

In plain language: a time-orientable spacetime admits two opposite global orientations. The projected extensions distinguish one of them if they form a non-empty set of timelike directions that all agree with that orientation. Under reversal, those same directions point the opposite way.

Proof. The only other global time orientation is $-\tau$. Every timelike vector future-directed relative to τ is past-directed relative to $-\tau$. Since S is non-empty, at least one projected outgoing history-direction exists. Therefore $-\tau$ is incompatible with the requirement that projected extension-directions be future-directed. $\square$

What the proof establishes: Exactly one global orientation agrees with the projected extension order under the stated conditions. Applying this result to a physical universe requires an independently justified extension structure and projection satisfying those conditions.

The result is a compatibility test between two independently specified structures: the extension order in E and the time orientations available on M . Its explanatory content depends on how the order and projection are physically justified. The following failure cases make that requirement explicit.

The physically substantive work lies in Π . In the present architecture, Π is a structurally specified projection from physically admissible modal configurations to M ; it is not generated by a variational evaluation or by the selection of an actual history. The proposal fails if all outgoing history-directions are collapsed, if their images are spacelike, or if different projected directions demand opposite orientations. It also fails if M is not time-orientable. A modal asymmetry therefore does not automatically become a temporal one.

Global non-emptiness of projected directions at every point is not required for logical uniqueness. On a connected time-orientable spacetime, once the projected structure is non-empty and globally compatible with one orientation, the opposite orientation is excluded. Wider coverage matters for explanatory robustness: a proposal in which Π carries orientation only in an isolated patch would be much less attractive than one in which the same alignment is stable throughout the physically relevant domain. Having established when a global orientation is selected, the next section separates that result from the thermodynamic arrow.

6 Orientation is not the thermodynamic arrow

A temporal orientation consistently designates one family of timelike directions as future-directed throughout spacetime. Entropy growth is a further property of physical processes. This section explains their relation and why fixing orientation independently of entropy makes their alignment a substantive physical question.

An orientation assigns the labels future-directed and past-directed to the two families of timelike directions. Entropy describes the physical states encountered along a history. Relative to a fixed orientation, entropy can increase, decrease or remain constant, depending on preparation and dynamics.

This point also limits Proposition 1. Our actual universe contains thermodynamic asymmetry. If that asymmetry is admitted into D , reversal invariance fails and the no-go does not apply. The paper therefore does not claim that modal structure is needed to read the orientation off the actual entropy gradient. It asks whether orientation can be grounded independently of that gradient.

The distinction matters for explanatory order. If future is defined as the direction of entropy increase, then one cannot subsequently explain why entropy increases toward the future without circularity. An independently fixed orientation makes a stronger question available: why does a special low-volume preparation relax in the direction already selected by the modal-spacetime bridge?

Explaining thermodynamic growth requires a physical account of preparation, the measure defining macrostate volumes and the dynamics governing relaxation. These are additional inputs to the orientation construction. Even a weighted, irregular branching structure can coexist with reversible dynamics.

7 When temporal and thermodynamic arrows come apart

The distinction can be tested in a block whose entropy is lowest in the middle and grows toward both ends. We compare a modal root at one edge with a root at the central preparation. An edge root can support one global orientation despite opposing local entropy arrows. A central root whose extensions project outward toward both ends fails the global compatibility test.

The distinction between temporal orientation and thermodynamic irreversibility becomes substantive only if the two can come apart. Cosmological models with an interior low-complexity or low-entropy region provide the natural test. Janus-point models and two-arrow cosmologies have been discussed independently in physics (Barbour, Koslowski, and Mercati 2014; Carroll and Chen 2004). The present argument uses a minimal toy dynamics to isolate the orientation issue.

Let M have a global coordinate $t \in [-T, T]$, without using the sign of t to define the future. Place a low-entropy preparation at $t=0$. Take $p_0=(1,0)$, equilibrium $\pi=(1/2,1/2)$, and the doubly

stochastic, detailed-balance kernel $K = \begin{bmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{bmatrix}$. Define $p(t) = K^{|t|} p_0$. Its macro-entropy is symmetric: $S(0)=0$, $S(\pm 1)=0.562335\dots$, $S(\pm 2)=0.661563\dots$, and $S(t)$ tends to $\log 2$ as $|t|$ increases. The thermodynamic arrows therefore point away from the central preparation on both sides.

What does the modal account say? The answer depends on a structural fact that entropy does not determine: the location of the root of E relative to the physical preparation. Two cases must be distinguished.

Case 1: edge root. Suppose the minimal root of E projects to the $t=-T$ side and all projected outgoing history-directions are compatible with $\tau=+\partial t$. Proposition 2 then selects one global temporal orientation. The rare low-entropy preparation at $t=0$ is a downstream physical preparation, not the modal root. Relative to τ , entropy decreases as the history approaches $t=0$ from the left and increases after $t=0$. Hence one global temporal orientation coexists with two oppositely directed local thermodynamic arrows.

Case 2: Janus root. Suppose instead that the modal root is identified with the central preparation and outgoing extensions project toward $-\partial t$ on one side and $+\partial t$ on the other. Then the hypotheses of Proposition 2 fail globally: no single time orientation makes all projected outgoing directions future-directed. The modal account therefore selects no global orientation by Proposition 2. It supports only the two local outward orientations supplied by the projected extension structure.

The contrast is useful because neither verdict is obtained by defining the future as the direction of entropy increase. In the edge-root case the modal and thermodynamic accounts disagree on the left half of the block. In the Janus-root case the global bridge fails for a structural reason: the projected extension order itself points both ways from the root. The location of the modal root, not the sign of the entropy gradient, determines which modal verdict obtains.

The difference in the edge-root case is presently metaphysical rather than observational. Both descriptions agree on the physical entropy profile and other block-level facts; they disagree on which direction is called future on the left side. The example nevertheless shows that the separation between temporal orientation and thermodynamic arrow is not merely a change of notation inside the theory: the two principles assign different orientation structure to the same entropy profile.

The example also forces a useful distinction inside the broader AE programme. The modal root r_E and a thermodynamic low preparation P_{low} are not constitutively identical. Earlier root-preparation models instantiate the special case in which they coincide. The current recording mechanism concerns a task-constrained local preparation; it does not identify that preparation with the modal root (Mury 2026d).

The remaining pressure point concerns material records. The abstract record result is information-theoretic rather than thermodynamic. Formally, record information can accumulate along an extension order even where Boltzmann entropy decreases relative to a global orientation. Whether memories, fossils, radiation records and other material traces can

actually remain aligned with that order on the thermodynamically opposed side of a Janus block is an open physical test. Failure would limit the physical-record unification claim, not Proposition 2 itself. The argument can now be placed within the wider AE architecture without allowing its later components to enter the orientation proof.

8 Arborescent Eternalism as one implementation

Sections 2–7 establish the orientation module. In AE, its inputs are E and Π : the rooted extension order and its physical projection. Reference weights, global evaluation and actuality operate on the histories represented within that architecture. This section explains their relation to the orientation module.

Arborescent Eternalism (AE) is one implementation of the architecture described above (Mury 2026a). It combines a tenseless physical block M with a structured domain E of admissible histories and a structurally specified projection Π from the physically admissible part of E to M . Histories are complete; branching represents objective or nomological alternatives rather than events waiting to become real. In the broader AE architecture, global evaluation and selection operate only after E , Π and the reference-weight structure have been specified; they do not produce Π .

A global evaluator of the form $J(H \mid E, \Pi, w, A) = -\log w_E(H) + B(H) + C_A(H)$, where w_E is the fixed reference weight, B a non-negative physical cost and C_A a goal-compatibility term, respects that order of dependence. It ranks already admissible complete histories and may select H^* provisionally by minimisation. It does not supply the rooted extension order, construct Π , or choose between the two spacetime orientations. Consequently, a change in A may change which complete history is selected while leaving the orientation result untouched: Proposition 2 depends on E and Π , not on the identity of the minimiser.

The adjective 'arborescent' should not be read as a commitment to a strict graph-theoretic tree. E may contain independent events and compatible reconvergences. What matters for the orientation argument is the rooted antecedence-to-extension structure and the preservation of provenance for incompatible histories.

AE combines two ideas without identifying them. From eternalism it keeps tenseless completeness. From branching approaches it keeps objective modal structure. Proposition 2 supplies the missing bridge: modal orientation and temporal orientation are distinct, and their alignment succeeds only under explicit projection conditions.

This is the sense in which E can be an explanatory enrichment of a B-theoretic block. E is not logically required for eternalism, and the paper does not claim that every B-theory must contain this particular ontology. Rather, a reversal-symmetric B-theory that seeks a modal explanation of orientation must add some asymmetric structure capable of doing equivalent work. AE proposes to use the same structure already employed for alternatives and records.

The comparison remains open-ended. A geometrical asymmetry in M may be simpler in some theories. A primitive orientation may be acceptable to a primitivist. The case for AE

becomes stronger only insofar as the same E and Π also constrain other phenomena without adding independent primitives for each. This paper establishes the orientation component of that broader unification claim, not the whole programme.

9 Comparative map: four ways of obtaining directedness

AE is not the only way to supply the asymmetry demanded by Proposition 1. This section compares primitive direction, asymmetric geometry, open-future branching and tenseless modal directedness by asking where each places the asymmetry and what else that structure explains. The comparison prepares the test of whether AE genuinely improves on a primitive temporal orientation.

The debate is easier to assess if four strategies are kept apart. The first is primitive direction. A theory can take one temporal orientation as part of the fundamental structure. This is metaphysically straightforward: no reduction is promised, so no reduction fails. The cost is explanatory only if one expected direction to follow from other independently motivated structure. Maudlin's defence of irreducible temporal structure makes this option philosophically serious rather than a fallback.

The second strategy is geometric. A spacetime may possess global features that are not invariant under reversal, and these features can orient local physical quantities. Castagnino and Lombardi's non-entropic programme is the clearest comparator. Their arrow is grounded in global geometrical asymmetry and connected to local energy flow. Relative to Proposition 1, this strategy rejects $\Theta D = D$ by enriching D with asymmetric geometry.

The third strategy is open-future branching. Here the asymmetry is modal. Histories share what is settled and diverge with respect to what remains possible. This has an explanatory virtue absent from a bare primitive orientation: the same structure that represents objective possibility also distinguishes directions. But it raises the question whether the asymmetry reflects an ontologically open future, an A-theoretic ingredient, or an independently justified feature of physical possibility.

The fourth strategy is tenseless modal directedness. It keeps the unificatory ambition of branching while dropping the claim that one temporal side is ontologically unsettled. The extension order is asymmetric, but every admissible history can be complete. Temporal orientation is not identified with modal extension; it is selected only when a separate physical bridge aligns extension-directions with one of the two orientations of M.

These strategies are not ordered by a simple scale of metaphysical virtue. Primitive direction is formally cheaper than a large modal structure. Geometrical asymmetry may be more directly physical. Open-future branching may be attractive to an indeterminist who already accepts a settled/unsettled distinction. The present proposal is attractive specifically when one wants full tenselessness, objective modality, and a non-primitive orientation together.

Eternalist branching models occupy an intermediate position. They show that branching semantics need not imply presentism or a growing block. But a tenseless formulation can retain a fixed-past asymmetry. The question then becomes whether that asymmetry is stipulated or connected to independently defined physical structure. The projection approach makes that connection explicit and exposes its failure modes.

Farr's criticism is relevant here. If a branching formalism simply prohibits backward branching, it may encode a time-asymmetric modal structure without explaining why physical possibility has that form. The present proposal does not eliminate that burden. It relocates it to two visible places: the primitive root-to-extension asymmetry in E and the bridge Π .

The comparison with C-theories is equally instructive. A C-theorist may deny that the two orientations correspond to a fundamental physical fact. Nothing in the mathematics forces realism about direction. The modal account is conditional realism: if one wants an objective direction that is not defined thermodynamically and not inserted directly into M , a tenseless modal structure provides one coherent route.

Finally, Proposition 1 should not be mistaken for the claim that symmetric laws can never coexist with asymmetric phenomena. Spontaneous symmetry breaking is the obvious warning. The claim is narrower: an intrinsic equivariant rule evaluated on exactly symmetry-invariant data cannot return one member of a pair exchanged by that symmetry. Asymmetric states, boundaries or stochastic selection are additional explanatory resources.

10 The unification test for the modal primitive

The main explanatory objection is that AE may replace a primitive temporal orientation with a more elaborate primitive modal order. That replacement adds value only if the same independently specified structure also constrains other phenomena. This section asks whether the structures used for orientation can also represent alternatives, records, weights and the domain of global selection without being adjusted separately for each task.

The primitive-exchange objection can now be made precise. Let T contain a primitive time orientation on M . Let T^+ replace it with a rooted modal order E plus Π . If E and Π do nothing except recover the orientation already available in T , T^+ is not explanatorily better. A genuine gain requires reuse: the same structure must be independently motivated and constrain several targets.

The first reuse is constitutive. E represents alternatives. Maximal coherent histories and finite configurations distinguish compatibility, incompatibility and provenance. This is not an orientation-specific construction. If objective modality is wanted for counterfactual or nomological reasons, a structured history space is motivated before the orientation question is asked.

The second reuse is record structure. If X labels alternatives and a physical record $Y=R(X)$ is deterministic with at most m antecedents per record, then $\text{Ent}(Y) \geq \text{Ent}(X) - \log m$. An injective record map preserves branch information exactly. If later records deterministically

retain earlier records, record information cannot decrease along the extension order. This is not Boltzmann entropy; it is a conditional result about trace fidelity.

The third reuse is probabilistic and evaluative. Once histories and genuine cylinders are defined on E, fixed reference weights and global selection rules can act on the same already specified domain. The current companion entropy paper studies a local recording mechanism connecting task-constrained preparation to typical macroscopic growth; it distinguishes that result from global selection. Nothing in the orientation argument depends on that result (Mury 2026d).

The established contributions are the conditional orientation result and the record-information bound. The same modal domain also supports reference weights and global evaluation. The unification hypothesis is that one independently justified structure can sustain these roles together; its physical realisation remains a research obligation.

AE is itself an enrichment of a B-theoretic block. A faithful formulation in B-theoretic language must retain the rooted modal order and projection used for orientation, the trace relations used for records, and the history space and measures used for weighting. Expressing these structures in another vocabulary does not provide a competing explanation. The remaining questions concern their objective or instrumental interpretation and whether their common use yields an explanatory gain. Nothing here establishes empirical distinctiveness.

The relevant standard of parsimony is how much explanatory work the structure performs. A primitive orientation suffices for a theory seeking only a temporal direction. A common extension structure becomes attractive when a theory also needs objective alternatives, ordered records and a domain for weighting and selection.

This unification criterion is also a research constraint. Future additions count as support only if they are derived from, or independently constrained by, the same structures already used elsewhere. A low-entropy boundary cannot simply be hidden inside a weight or projection chosen after the fact. Otherwise the apparent unification is notation rather than explanation. The objections in the next section test the limits of precisely this claim.

11 Objections

The objections distinguish three claims: that exactly reversal-symmetric explanatory data cannot distinguish an orientation, that rooted modal structure is the only possible source of orientation, and that the result already explains thermodynamic irreversibility. The paper proves the first under the stated symmetry assumptions and establishes a conditional route through modal structure. It does not defend the latter two claims.

11.1 The asymmetry has only been moved

Correct. The extension orientation is primitive. Proposition 1 explains why a preferred orientation cannot be obtained from perfectly reversal-symmetric data by an intrinsic equivariant rule. The proposal therefore relocates asymmetry into the rooted modal structure

rather than deriving it from nothing. The potential gain is that this structure is also independently used to represent branching possibilities in a tenseless world. Its explanatory value consequently depends on whether those modal and further record-related roles survive independent scrutiny. Section 10 states that unification test.

11.2 A boundary condition can orient the block

Correct. A sufficiently asymmetric boundary condition in M can evade Proposition 1 and orient the block. The present paper does not prove E uniquely necessary. It offers a different explanatory location for the asymmetry and makes the comparison explicit.

11.3 Branching is unnecessary for orientation

Also correct. A single rooted chain whose projected direction is timelike could satisfy Proposition 2. Branching matters because the intended modal structure represents objective alternatives. The result is therefore not that time requires a tree. It is that branching-style modal directedness can survive the removal of an open future.

11.4 Eternalist branching already exists

Yes. The novelty is not eternalism plus branching. The contribution is to separate a tenseless modal extension order from spacetime orientation and state the bridge conditions under which the former selects the latter.

11.5 The actual universe is not reversal-symmetric

Probably not, once thermodynamic and cosmological asymmetries are included. Proposition 1 is explicitly conditional on the chosen data D. Its purpose is to isolate the explanatory problem before contingent arrows are used. It therefore cannot be used as an argument that the actual block, with all physical facts included, is unoriented.

11.6 The proposal is not yet a theory of the arrow of time

Agreed if 'arrow of time' means thermodynamic irreversibility. The paper concerns temporal directedness or orientation. Entropy, low-entropy preparation and relaxation require a further physical bridge.

12 Conclusion

Eternalism and branching need not be rival pictures. Eternalism says that complete histories can be tenseless. Branching says that physical possibilities can possess objective structure. The tension arises when the explanatory force of branching is tied essentially to an ontologically open future.

The argument separates that force from open futurism. Relative to reversal-symmetric data, a block cannot intrinsically choose one of its two time orientations. A rooted antecedence-to-extension relation among complete histories can supply an asymmetric modal structure.

When the projected history-directions are timelike and globally compatible with one of the two orientations of a connected time-orientable block, that orientation is uniquely selected relative to the extension order.

The Janus example makes the separation non-verbal, but its verdict depends on the relation between the modal root and the low-entropy preparation. With an edge root, E and Π can select one global orientation while entropy decreases toward an interior minimum on one side and increases away from it on the other. With a root at the central preparation, the projected extensions point outward in both coordinate directions, so the global compatibility condition of Proposition 2 fails and only two local outward orientations are obtained. In neither case is temporal orientation defined by the sign of the entropy gradient.

The result establishes a conditional route from tenseless modal directedness to physical orientation. Its assumptions locate the asymmetry in the extension order and require a compatible projection. The local recording mechanism connecting task-constrained preparation with typical macroscopic growth is developed separately (Mury 2026d).

The proposed explanatory gain is reuse. The extension structure represents alternatives, supports conditional accumulation of record information, and supplies the domain for weighting and selection. Demonstrating these roles in a common physical realisation would substantiate AE's unification claim.

Arborescent Eternalism develops this strategy as a common architecture for possibility, temporal orientation and physical processes within a tenseless block. The present paper establishes the orientation step under explicit bridge conditions. The programme then asks how preparation and relaxation align thermodynamic growth with the orientation fixed by that structure. This order of explanation keeps the alignment open to physical investigation.

Declaration on the use of generative AI

Generative AI systems, including OpenAI's ChatGPT and Codex and Anthropic's Claude, were used during the development of this manuscript for conceptual exploration, adversarial review, formalisation, drafting and structural revision, and to assist with literature discovery and bibliographic verification. The symmetry argument, bridge conditions, philosophical interpretation and final editorial decisions are those of the author. No AI system is credited as an author or treated as an evidential source. All AI-assisted material retained in the manuscript was checked, revised and approved by the author, who accepts full responsibility for the arguments, calculations, citations and final text.

AI-assisted work was organised through a research harness tracking assumptions, experiments, evidence and decisions. Numerical results were produced by executable Python scripts, and arguments underwent adversarial review; responsibility for the submitted claims remains with the author.

Proof and source availability

The proofs are given in the text. Their dependencies and scope are documented in the online proof repository:

https://github.com/Legiolab/arborescent_eternalism_public/tree/main/proofs/paper-a-temporal-directedness/

This archive contains only proof-facing material; private development harnesses, governance files and exploratory logs are not part of the evidential basis.

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